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Experiment 1 :Data Preprocessing and Cleaning

Learning Objectives

The objective of this experiment is to understand the importance of data preprocessing in data science and machine learning. It involves learning how to load and explore a customer churn dataset, identify and handle missing values and duplicate records, convert data into appropriate formats, encode categorical variables, standardize numerical features, and visualize important patterns in the data. These preprocessing techniques help prepare a clean and reliable dataset for building accurate machine learning models.

Methods Used / Python Libraries Involved and Purpose

- **Pandas** – Used to load, explore, clean, and manipulate the customer churn dataset using DataFrames. □ **NumPy** – Used to perform numerical operations and identify numerical columns for preprocessing.
- **Matplotlib.pyplot** – Used to create visualizations such as histograms and other plots for data analysis.
- **Seaborn** – Used to generate statistical visualizations including distribution plots, count plots, box plots, and correlation heatmaps.
- **LabelEncoder (Scikit-learn)** – Used to convert categorical variables into numerical values so they can be processed by machine learning algorithms.
- **StandardScaler (Scikit-learn)** – Used to standardize numerical features by scaling them to have a mean of 0 and a standard deviation of 1, improving model performance.

Program code

```
# 1. Import required Python libraries
import   matplotlib.pyplot   as   plt
import numpy as np import pandas as
pd import seaborn as sns
from sklearn.preprocessing import LabelEncoder, StandardScaler

url = 'https://raw.githubusercontent.com/alexeygrigorev/mlbookcamp-code/master/chapter-03-
churnprediction/WA_Fn-UseC_-Telco-Customer-Churn.csv' df = pd.read_csv(url)

print('--- First 10 records ---
')
display(df.head(10))

print('--- Last 10 records ---')
display(df.tail(10))

print('--- Shape of the dataset ---', df.shape)
print('--- Column names ---', df.columns.tolist())


# 3. Explore the Dataset print('---
Dataset Info ---') df.info()

print('--- Summary Statistics ---
')
display(df.describe())

print('--- Number of missing values per column ---')
print(df.isnull().sum())
print('--- Number of duplicate records ---', df.duplicated().sum())
```

```
# 4. Clean the data
# a. Remove duplicate customer records df.drop_duplicates(inplace=True)


# c. Convert the TotalCharges column into numeric datatype (coercing errors to NaN)
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce') # b. Replace
missing values # i. Numerical columns -> Median num_cols =
df.select_dtypes(include=[np.number]).columns for
col in num_cols:
  df[col].fillna(df[col].median(), inplace=True)


# ii. Categorical columns -> Mode cat_cols =
df.select_dtypes(include=['object']).columns for
col in cat_cols:
  df[col].fillna(df[col].mode()[0], inplace=True)


# d. Verify that no missing values remain print(
    '--- Missing values after cleaning ---',
df.isnull().sum().sum(),
)


# 5. Encode categorical columns: Gender, Partner, Dependents, Contract, PaymentMethod,
InternetService, Churn target_cat_cols = [
    'gender',
    'Partner',
    'Dependents',
    'Contract',
    'PaymentMethod',
    'InternetService',
    'Churn',
]
# Filter to ensure columns exist in dataframe target_cat_cols =
[c for c in target_cat_cols if c in df.columns]

le = LabelEncoder() for col
in target_cat_cols:
  df[col] = le.fit_transform(df[col].astype(str))
 # 6. Standardise the numerical features: Tenure, MonthlyCharges, TotalCharges, using
StandardScaler
scale_cols = ['tenure', 'MonthlyCharges', 'TotalCharges']
scale_cols = [c for c in scale_cols if c in df.columns]

scaler = StandardScaler() df[scale_cols] =
scaler.fit_transform(df[scale_cols])


# 7. Plot Distribution of MonthlyCharges, Distribution of Tenure
plt.figure(figsize=(12, 5)) plt.subplot(1, 2, 1)
sns.histplot(df['MonthlyCharges'], kde=True, color='blue') plt.title('Distribution
of MonthlyCharges')
```

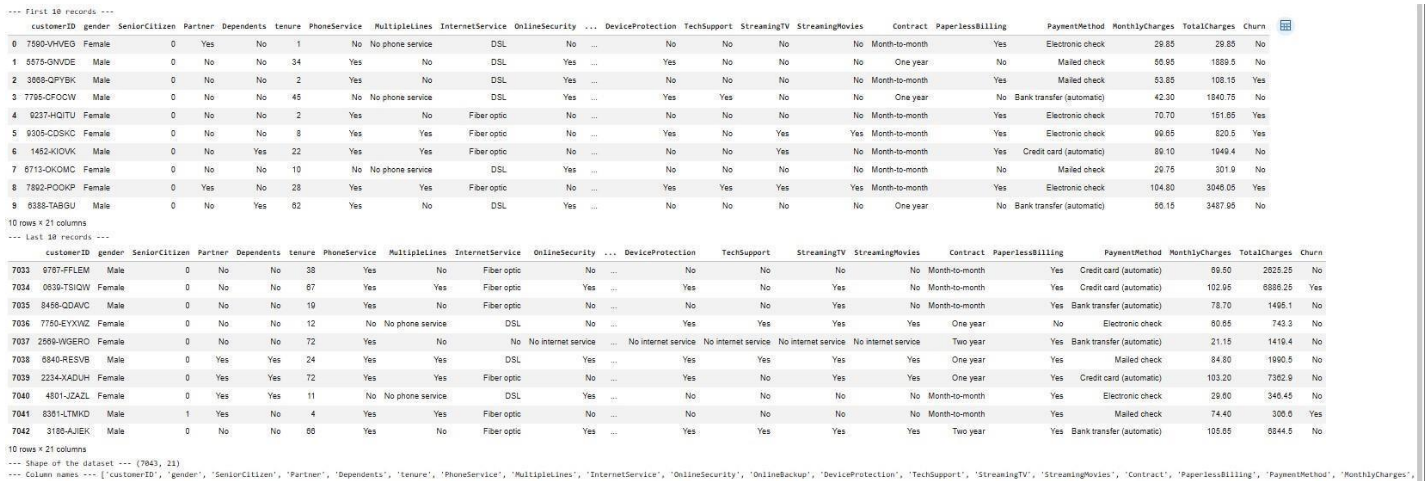
```
plt.subplot(1,   2,   2)   sns.histplot(df['tenure'],
kde=True, color='green') plt.title('Distribution of
Tenure') plt.tight_layout() plt.show()
```

```
# 8. Create a Count Plot showing Contract vs Churn
plt.figure(figsize=(6, 4))
sns.countplot(x='Contract', hue='Churn', data=df)
plt.title('Contract vs Churn') plt.show()
```

```
# 9. Draw a Box Plot of MonthlyCharges grouped by Churn
plt.figure(figsize=(6,    4))    sns.boxplot(x='Churn',
y='MonthlyCharges', data=df) plt.title('MonthlyCharges
grouped by Churn') plt.show()
```

```
#    10.    Display    the    Correlation    Heatmap    of    all    numerical    features
plt.figure(figsize=(8, 6))
num_df            =            df.select_dtypes(include=[np.number])
sns.heatmap(num_df.corr(), annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Heatmap') plt.show()
```

Output



--- First 10 records ---

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	No	No	No	No	Month-to-month	Yes	Electronic check	29.85	29.85	No
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	Yes	No	No	No	One year	No	Mailed check	56.95	1889.5	No
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	No	No	No	No	Month-to-month	Yes	Mailed check	53.85	108.15	Yes
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...	Yes	Yes	No	No	One year	No	Bank transfer (automatic)	42.30	1840.75	No
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	...	No	No	No	No	Month-to-month	Yes	Electronic check	70.70	151.65	Yes
5	9305-CDSKC	Female	0	No	No	8	Yes	Yes	Fiber optic	No	...	Yes	No	Yes	Yes	Month-to-month	Yes	Electronic check	99.65	820.5	Yes
6	1452-KIOVK	Male	0	No	Yes	22	Yes	Yes	Fiber optic	No	...	No	No	Yes	No	Month-to-month	Yes	Credit card (automatic)	89.10	1949.4	No
7	6713-OKOMC	Female	0	No	No	10	No	No phone service	DSL	Yes	...	No	No	No	No	Month-to-month	No	Mailed check	29.75	301.9	No
8	7892-POOKP	Female	0	Yes	No	28	Yes	Yes	Fiber optic	No	...	Yes	Yes	Yes	Yes	Month-to-month	Yes	Electronic check	104.80	3046.05	Yes
9	6388-TABGU	Male	0	No	Yes	62	Yes	No	DSL	Yes	...	No	No	No	No	One year	No	Bank transfer (automatic)	56.15	3487.95	No

10 rows × 21 columns

--- Last 10 records ---

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
7033	9767-FFLEM	Male	0	No	No	38	Yes	No	Fiber optic	No	...	No	No	No	No	Month-to-month	Yes	Credit card (automatic)	69.50	2625.25	No
7034	0639-TSIQW	Female	0	No	No	67	Yes	Yes	Fiber optic	Yes	...	Yes	No	Yes	No	Month-to-month	Yes	Credit card (automatic)	102.95	6886.25	Yes
7035	8456-QDAVC	Male	0	No	No	19	Yes	No	Fiber optic	No	...	No	No	Yes	No	Month-to-month	Yes	Bank transfer (automatic)	78.70	1495.1	No
7036	7750-EYXWZ	Female	0	No	No	12	No	No phone service	DSL	No	...	Yes	Yes	Yes	Yes	One year	No	Electronic check	60.65	743.3	No
7037	2569-WGERO	Female	0	No	No	72	Yes	No	No	No internet service	...	No internet service	No internet service	No internet service	No internet service	Two year	Yes	Bank transfer (automatic)	21.15	1419.4	No
7038	6840-RESVB	Male	0	Yes	Yes	24	Yes	Yes	DSL	Yes	...	Yes	Yes	Yes	Yes	One year	Yes	Mailed check	84.80	1990.5	No
7039	2234-XADUH	Female	0	Yes	Yes	72	Yes	Yes	Fiber optic	No	...	Yes	No	Yes	Yes	One year	Yes	Credit card (automatic)	103.20	7362.9	No
7040	4801-JZAZL	Female	0	Yes	Yes	11	No	No phone service	DSL	Yes	...	No	No	No	No	Month-to-month	Yes	Electronic check	29.60	346.45	No
7041	8361-LTMKD	Male	1	Yes	No	4	Yes	Yes	Fiber optic	No	...	No	No	No	No	Month-to-month	Yes	Mailed check	74.40	306.6	Yes
7042	3186-AJIEK	Male	0	No	No	66	Yes	No	Fiber optic	Yes	...	Yes	Yes	Yes	Yes	Two year	Yes	Bank transfer (automatic)	105.65	6844.5	No

10 rows × 21 columns

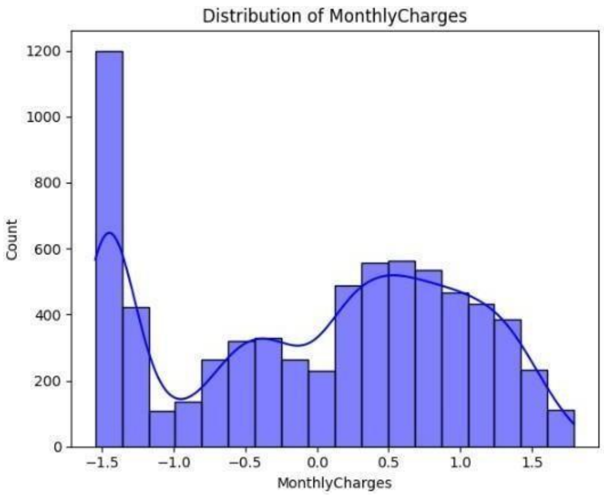
```
--- Shape of the dataset --- (7043, 21)
--- Column names --- ['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure', 'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod', 'MonthlyCharges',
```

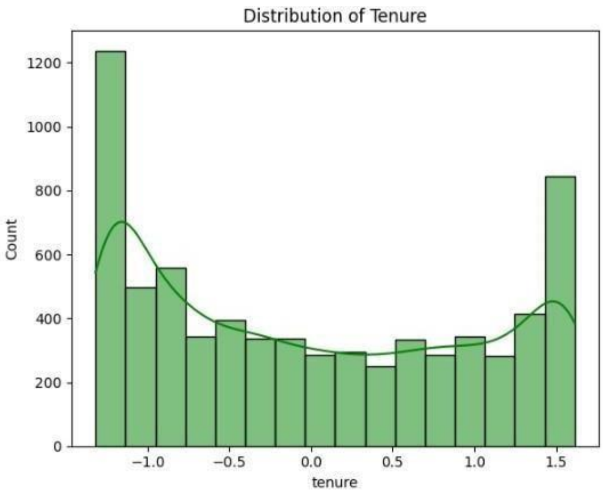
```
--- Dataset Info ---
<class 'pandas.core.frame.DataFrame'> RangeIndex:
7043 entries, 0 to 7042
Data columns (total 21 columns):
  # Column              Non-Null Count Dtype


........  ...............          ......................................   ...........



0   customerID        7043 non-null object
1   gender    7043 non-null object
2   SeniorCitizen     7043 non-null int64
3   Partner   7043 non-null object
4   Dependents        7043 non-null object
5   tenure    7043 non-null int64
6   PhoneService      7043 non-null object
7   MultipleLines     7043 non-null object
8   InternetService 7043 non-null object
9   OnlineSecurity    7043 non-null object
10  OnlineBackup      7043 non-null object
11  DeviceProtection 7043 non-null object
```

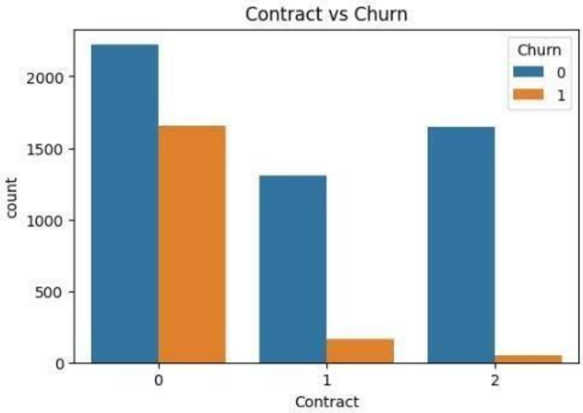
```
12  TechSupport      7043 non-null object
13  StreamingTV      7043 non-null object
14  StreamingMovies 7043 non-null object
15  Contract  7043 non-null object
16  PaperlessBilling 7043 non-null object
17  PaymentMethod    7043 non-null object


18 MonthlyCharges     7043 non-null   float64
19 TotalCharges       7043 non-null   object 20
    Churn             7043 non-null   object
    dtypes: float64(1), int64(2), object(18)
    memory usage: 1.1+ MB --- Summary Statistics
    ---
```

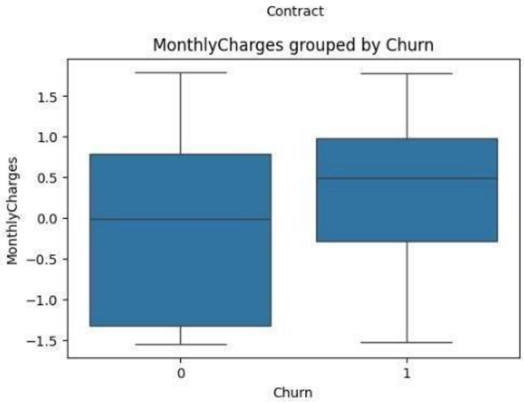
	SeniorCitizen	**tenure**	**MonthlyCharges**
count	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692
std	0.368612	24.559481	30.090047
min	0.000000	0.000000	18.250000
25%	0.000000	9.000000	35.500000
50%	0.000000	29.000000	70.350000
75%	0.000000	55.000000	89.850000
max	1.000000	72.000000	118.750000

```
--- Number of missing values per column ---
customerID          0 gender              0
SeniorCitizen       0
Partner             0
Dependents          0 tenure
0 PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0 Churn
0 dtype: int64
--- Number of duplicate records----- 0
--- Missing values after cleaning ----- 0
```

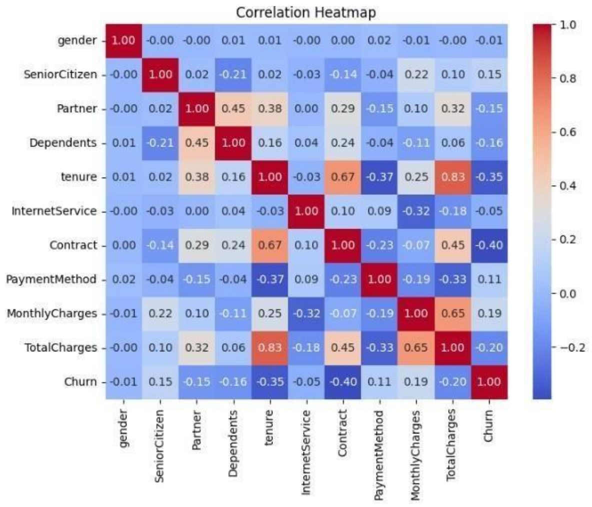

Distribution of MonthlyCharges


Distribution of Tenure


Contract vs Churn


Contract

MonthlyCharges grouped by Churn


Correlation Heatmap

	gender	SeniorCitizen	Partner	Dependents	tenure	InternetService	Contract	PaymentMethod	MonthlyCharges	TotalCharges	Churn
gender	1.00	-0.00	-0.00	0.01	0.01	-0.00	0.00	0.02	-0.01	-0.00	-0.01
SeniorCitizen	-0.00	1.00	0.02	-0.21	0.02	-0.03	-0.14	-0.04	0.22	0.10	0.15
Partner	-0.00	0.02	1.00	0.45	0.38	0.00	0.29	-0.15	0.10	0.32	-0.15
Dependents	0.01	-0.21	0.45	1.00	0.16	0.04	0.24	-0.04	-0.11	0.06	-0.16
tenure	0.01	0.02	0.38	0.16	1.00	-0.03	0.67	-0.37	0.25	0.83	-0.35
InternetService	-0.00	-0.03	0.00	0.04	-0.03	1.00	0.10	0.09	-0.32	-0.18	-0.05
Contract	0.00	-0.14	0.29	0.24	0.67	0.10	1.00	-0.23	-0.07	0.45	-0.40
PaymentMethod	0.02	-0.04	-0.15	-0.04	-0.37	0.09	-0.23	1.00	-0.19	-0.33	0.11
MonthlyCharges	-0.01	0.22	0.10	-0.11	0.25	-0.32	-0.07	-0.19	1.00	0.65	0.19
TotalCharges	-0.00	0.10	0.32	0.06	0.83	-0.18	0.45	-0.33	0.65	1.00	-0.20
Churn	-0.01	0.15	-0.15	-0.16	-0.35	-0.05	-0.40	0.11	0.19	-0.20	1.00

Learning outcome

The experiment provided an understanding of the complete data preprocessing workflow for a customer churn dataset. Skills were developed in loading and exploring datasets, identifying and handling missing values and duplicate records, converting data into appropriate formats, encoding categorical variables, standardizing numerical features, and creating meaningful visualizations. A clean, consistent, and machine learning-ready dataset was successfully prepared, establishing a strong foundation for predictive modeling and data analysis.